STAT 314500 - Applied Partial Differential Equations

Assignment 7: PDEs on multi-dimensional domains, Green's functions and Fourier transforms

This is an individual assignment. You may collaborate with others but are expected to submit your own work. Show all relevant work. Incomplete work will be docked points.

Note: if a question is marked with an asterisk, *, and appears vague - that's on purpose. Questions marked * are conceptual short-answer questions. Do your best to reason these through, and describe that reasoning concisely. We are looking more for reasoning than precision.

Reading: Chapters 7, 9 and 10 in Applied Partial Differential Equations with Fourier Series and Boundary Value Problems (Richard Haberman).

Question 1 Let the Green's function $G(x; x_0)$ satisfy

$$\begin{aligned}\frac{\mathrm{d}G}{\mathrm{d}x} + G &= \delta(x - x_0)\\ G(0; x_0) &= 0\,.\end{aligned}$$

Solve for this Green's function directly. Is this Green's function symmetric, i.e, is $G(x; x_0) = G(x_0; x)$? (10 points)

Question 2 Using the method of eigenfunction expansion, find the Green's function $G(x, y; x_0, y_0)$ that satisfies (10 points)

$$\begin{aligned}\Delta G &= \delta(x - x_0, y - y_0)\\ G(0, y; x_0, y_0) &= 0\\ G(L, y; x_0, y_0) &= 0\\ G(x, 0; x_0, y_0) &= 0\\ G(x, H; x_0, y_0) &= 0\,.\end{aligned}$$

Question 3 Find the Fourier transform of (10 points)

1. $$f(x) = \begin{cases} 1, & |x| < a \\ 0, & |x| > a\,. \end{cases}$$

2. $f(x) = e^{-a|x|}, a > 0.$

Question 4 Solve the diffusion equation with convection:

$$\begin{aligned}\partial_t u &= k\partial_{xx}u + c\partial_x u\\ u(x, 0) &= f(x)\,.\end{aligned}$$

Now suppose the initial condition is $\delta(x)$. Sketch the corresponding solution $u(x, t)$ for various values of t. What is the significance of the convection term in this setting? (10 points)

Question 5 Solve the heat equation

$$\frac{\partial u}{\partial t} = k\Delta u$$

in the following geometries. What can you say about the equilibrium temperature distribution?

1. Rectangular region $(x, y) \in [0, L] \times [0, H]$ subject to (5 points)

$$\begin{aligned}
&u(x, y, 0) = f(x, y) \\
&u(0, y, t) = 0 \\
&\frac{\partial u}{\partial x}(L, y, t) = 0 \\
&\frac{\partial u}{\partial y}(x, 0, t) = 0 \\
&\frac{\partial u}{\partial y}(x, H, t) = 0\,.
\end{aligned}$$

2. Semi-circular region $(r, \theta) \in [0, R] \times [0, \pi]$ with (5 points)

$$\begin{aligned}
u(r, \theta, 0) &= f(r, \theta) \\
u(r, 0, t) &= 0 \\
u(r, \pi, t) &= 0 \\
u(R, \theta, t) &= 0\,.
\end{aligned}$$

Question 1

We want $\frac{dG}{dx} + G = \delta(x - x_0)$, $G(0; x_0) = 0$

For $x \neq x_0$, we have $\frac{dG}{dx} + G = 0 \Rightarrow G(x; x_0) = Ce^{-x}$ on each side of $x = x_0 \Rightarrow G(x; x_0) = \begin{cases} Ae^{-x} & 0 < x < x_0 \\ Be^{-x} & x > x_0 \end{cases}$

Using the BC, $G(0; x_0) = A = 0 \Rightarrow G(x; x_0) = 0$ for $0 < x < x_0$.

Then at $x = x_0$, integrate across an interval around x_0: $\int_{x_0-\epsilon}^{x_0+\epsilon} [G_x + G]\,dx = \int_{x_0-\epsilon}^{x_0+\epsilon} \delta(x_0 - x)\,dx \Rightarrow G(x_0^+; x_0) - G(x_0^-; x_0) = 1$ then because $G(x_0^-; x_0) = 0$ we get $G(x_0^+; x_0) = 1$ and for $x > x_0$ $G(x; x_0) = Be^{-x}$

$\Rightarrow Be^{-x_0} = 1 \quad \therefore B = e^{x_0}$

So our function becomes $\boxed{G(x; x_0) = \begin{cases} 0 & 0 < x < x_0 \\ e^{x_0 - x} & x > x_0 \end{cases}}$ This is not symmetric because if $x < x_0$ then $G(x; x_0) = 0$ but $G(x_0; x) = e^{x - x_0} \neq 0$.

Question 2

The Dirichlet eigenfunction $\phi_{mn}(x, y) = \sin\left(\frac{m\pi x}{L}\right)\sin\left(\frac{n\pi y}{H}\right)$ $m, n \in \mathbb{N}$ which satisfies: $\Delta\phi_{mn} = -\left[\left(\frac{m\pi}{L}\right)^2 + \left(\frac{n\pi}{H}\right)^2\right]\phi_{mn}$ then $\lambda_{mn} = \left(\frac{m\pi}{L}\right)^2 + \left(\frac{n\pi}{H}\right)^2$

Then expanding the Green's function is eigenfunctions: $G(x, y; x_0, y_0) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} A_{mn}(x_0, y_0)\phi_{mn}(x, y)$ and the delta functions $\delta(x - x_0, y - y_0) = \frac{4}{LH}\sum_{m\geq 1}\sum_{n\geq 1} \phi_{mn}(x_0, y_0)\phi_{mn}(x, y)$ because $\int_0^L\int_0^H \phi_{mn}^2\,dy\,dx = \frac{LH}{4}$

Applying Δ to G gives $\Delta G = -\sum_{m\geq 1}\sum_{n\geq 1} \lambda_{mn} A_{mn}(x_0, y_0)\phi_{mn}(x, y)$

Then if we match coefficients with the coefficients of the delta expansion we get $-\lambda_{mn} A_{mn}(x_0, y_0) = \frac{4}{LH}\phi_{mn}(x_0, y_0) \Rightarrow A_{mn}(x_0, y_0) = -\frac{4}{LH}\frac{\phi_{mn}(x_0, y_0)}{\lambda_{mn}} \Rightarrow \boxed{G(x, y; x_0, y_0) = -\frac{4}{LH}\sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\frac{\sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{n\pi x_0}{L}\right)\sin\left(\frac{n\pi y}{H}\right)\sin\left(\frac{n\pi y_0}{H}\right)}{\left(\frac{m\pi}{L}\right)^2 + \left(\frac{n\pi}{H}\right)^2}}$

This satisfies the BC because the eigenfunction is zero when $x = 0$ or $L = 0$ or $y = 0$ or $y = H$

Question 3

Using $\hat{f}(\xi) = \int_{\mathbb{R}} f(x)e^{-i\xi x}\,dx$

1) let $f(x) = \begin{cases} 1 & |x| < a \\ 0 & |x| > a \end{cases} \Rightarrow \hat{f}(\xi) = \int_{-a}^{a} e^{-i\xi x}\,dx$ for $\xi \neq 0$ $\hat{f}(\xi) = \left[\frac{e^{-i\xi x}}{-i\xi}\right]_{-a}^{a} = \frac{e^{-i\xi a} - e^{i\xi a}}{-i\xi} = \frac{2\sin(a\xi)}{\xi}$

And at $\xi = 0$, $\hat{f}(0) = \int_{-a}^{a} 1\,dx = 2a \Rightarrow \boxed{\hat{f}(x) = \begin{cases} \frac{2\sin(a\xi)}{\xi} & \xi \neq 0 \\ 2a & \xi = 0 \end{cases}}$

2) Now $f(x) = e^{-a|x|}$, $a > 0$. $f(x)$ even $\Rightarrow \hat{f}(\xi) = \int_{-\infty}^{\infty} e^{-a|x|}e^{-i\xi x}\,dx = 2\int_0^{\infty} e^{-ax}\cos(\xi x)\,dx$ but then we can use the id $\int_0^{\infty} e^{-ax}\cos(\xi x)\,dx = \frac{a}{a^2 + \xi^2} \Rightarrow \boxed{\hat{f}(\xi) = \frac{2a}{a^2 + \xi^2}}$

Question 4

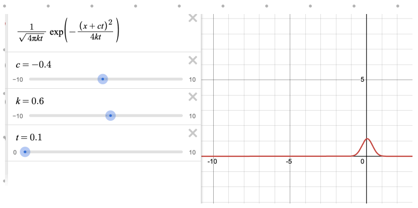
solve $u_t = Ku_{xx} + cu_x$, $u(x, 0) = f(x)$

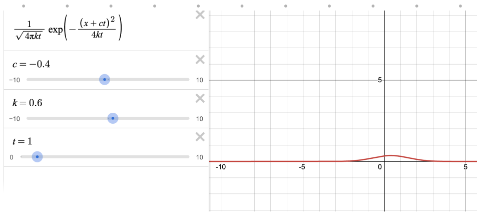
Then we take the fourier transform in x, then $\partial_t\hat{u}(\xi, t) = K(-\xi^2)\hat{u}(\xi, t) + c(i\xi)\hat{u}(\xi, t) \Rightarrow \partial_t\hat{u}(\xi, t) = (-K\xi^2 + ic\xi)\hat{u}(\xi, t)$

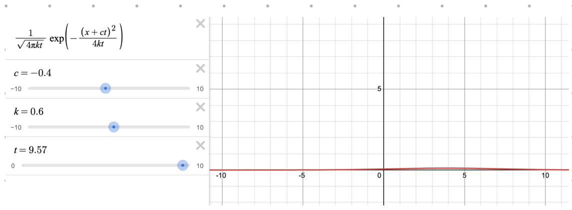
Then if we solve the ODE in t, we get $\hat{u}(\xi, t) = \hat{f}(\xi)e^{-K\xi^2 t + ic\xi t}$

$\Rightarrow u(x, t) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\hat{f}(\xi)e^{-K\xi^2 t + ic\xi t}e^{i\xi x}\,d\xi$. Then with the heat kernel: $\boxed{u(x, t) = \int_{-\infty}^{\infty}\frac{1}{\sqrt{4\pi Kt}}\exp\left(-\frac{(x + ct - y)^2}{4Kt}\right)f(y)\,dy \quad \text{for } t > 0}$

Now suppose the IC is $\delta(x)$. then all the initial mass is concentrated at $y = 0$ and since for delta functions: $\int_{-\infty}^{\infty}\phi(y)\delta(y)\,dy = \phi(0) \Rightarrow u(x, t)$ becomes $u(x, t) = \frac{1}{\sqrt{4\pi Kt}}\exp\left(-\frac{(x + ct)^2}{4Kt}\right)$







The convection term cu_x adds translation to the diffusion and transports it with velocity $-c$

Question 5

1) rectangular region with BCs = $u(0, y, t) = 0$, $\frac{\partial u}{\partial x}(L, y, t) = 0$, $\frac{\partial u}{\partial y}(x, 0, t) = 0$, $\frac{\partial u}{\partial y}(x, H, t) = 0$

Seperate solutions: $u(x, y, t) = X(x)\,Y(y)\,T(t)$

- X: into heat eq: $XYT' = K(X''YT + XY''T) \Rightarrow \frac{T'}{KT} = \frac{X''}{X} + \frac{Y''}{Y}$ so each spatial must contribute a constant $\Rightarrow \frac{X''}{X} = -\alpha^2$

  $\Rightarrow X'' + \alpha^2 X = 0$, $X(0) = 0$ & $X'(L) = 0 \Rightarrow X_n(x) = \sin(\alpha_n x)$, $\alpha_n = \frac{(n + \frac{1}{2})\pi}{L}$ $n \in \mathbb{N} \cup \{0\}$

- Y: the Y problem can similarly be broken up as $Y'' + \beta^2 Y = 0$. $Y'(0) = Y'(H) = 0 \Rightarrow Y_m(y) = \cos(\beta_m y)$, $\beta_m = \frac{m\pi}{H}$ $m \in \mathbb{N} \cup \{0\}$

- T: $T' = -K(\alpha_n^2 + \beta_m^2)T \Rightarrow T_{nm}(t) = e^{-K(\alpha_n^2 + \beta_m^2)t}$

$\therefore \boxed{u(x, y, t) = \sum_{n\geq 0}\sum_{m\geq 0} A_{nm}\sin\left(\frac{(n + 1/2)\pi x}{L}\right)\cos\left(\frac{m\pi y}{H}\right)e^{-K\left[\left(\frac{(n+1/2)\pi}{L}\right)^2 + \left(\frac{m\pi}{H}\right)^2\right]t}}$ where A_{nm} satisfies the $t = 0$ $f(x, y)$ IC

Since eigenvals $\left(\frac{(n + 1/2)\pi}{L}\right)^2 + \left(\frac{m\pi}{H}\right)^2 > 0$ then the solution decays as $t \to \infty$ with the exponential $\Rightarrow$ the eq temp is $u(x, y) = 0$.

2) semi circle: in polar, $\Delta u = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta}$ and the BCs are $u(r, 0, t) = 0$, $u(r, \pi, t) = 0$, $u(R, \theta, t) = 0$

seperate solutions: $u(r, \theta, t) = P(r)\,Q(\theta)\,T(t)$

the idea is very similar to the rectangular one

Q: $Q'' + \mu Q = 0$ w/ $Q(0) = Q(\pi) = 0 \Rightarrow Q_n(\theta) = \sin(n\theta)$ $n \in \mathbb{N}$

P: If we plug everything back into the equation, the P solution becomes: $r^2P'' + rP' + (\lambda r^2 - n^2)P = 0$ Then at sol $r \geq 0$, $F(r) = J_n(\sqrt{\lambda}\, r)$

the BC at $u(R,\theta,t) = 0$ gives $J_n(\sqrt{\lambda} R) = 0$, so $\sqrt{\lambda} R$ must be one positive zeros of J_n. If $j_{n\ell}$ denotes the ℓ-th positive zero of J_n then

$\sqrt{\lambda} R = j_{n\ell} \Rightarrow \lambda_{n\ell} = \left(\frac{j_{n\ell}}{R}\right)^2$

T: $T' = -k\lambda_{n\ell} T \Rightarrow T_{n\ell}(t) = e^{-k(j_{n\ell}/R)^2 t}$

$\therefore \boxed{u(r,\theta,t) = \sum_{n\geq 1} \sum_{\ell \geq 1} B_{n\ell} J_n\left(\frac{j_{n\ell} r}{R}\right) \sin(n\theta)\, e^{-k(j_{n\ell}/R)^2 t}}$ where the coefficients are chosen to fit the BC

Since $\left(\frac{j_{n\ell}}{R}\right)^2 > 0$, the solution decays as $t \to \infty \Rightarrow$ at equilibrium $u(r,\theta) = 0$